Confidence Calibration Test Student Handout

TOK Cognitive Science Activity Bank • Knowledge and the Knower • 20-25 min

Category	Knowledge and the Knower	**Time**	20-25 min
Difficulty	Easy	**ID**	confidence-calibration-test

Big idea: Certainty and accuracy are not the same.

Student-Facing Prompt

Answer each question, then mark your confidence: 50%, 60%, 70%, 80%, 90%, or 100%.

Concrete Stimulus Packet

Student Prompt	Answer each question privately. After marking, calculate accuracy within each confidence band.	Do not allow lookup; the point is calibration, not trivia mastery.
Question 1	Which planet is closest to the Sun?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 2	What is the capital city of Canada?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 3	Which is heavier: a kilogram of feathers or a kilogram of steel?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 4	In which year did the Berlin Wall fall?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 5	What is the chemical symbol for potassium?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%

Actual Lesson Questions

#	Question for students
1	Which planet is closest to the Sun?
2	What is the capital city of Canada?
3	Which is heavier: a kilogram of feathers or a kilogram of steel?
4	In which year did the Berlin Wall fall?
5	What is the chemical symbol for potassium?
6	Which country has the longest coastline in the world?
7	How many bones are in the adult human body?
8	What is the only mammal capable of true sustained flight?
9	Which has more letters: the Greek alphabet or the modern English alphabet?
10	Without looking it up, what is the 19th digit after the decimal point in pi?

Ready-to-Use Examples

• Ten general knowledge questions with confidence bands from 50% to 100%.

• A mini quiz mixing easy, medium, and impossible questions so overconfidence becomes visible.

• A class calibration chart comparing confidence level with actual accuracy.

Student Task

Complete the quiz, mark confidence for every answer, then compare confidence bands with actual accuracy.

Activity Steps

1. Give students general knowledge questions. For each answer they also write a confidence level: 50%, 60%, 70%, 80%, 90% or 100%.

2. Mark the answers.

3. For each confidence band, calculate actual percentage correct.

4. Ask students whether they were overconfident, underconfident or well calibrated.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

• Knowledge question: Is certainty a reliable indicator of knowledge?

• What is the difference between confidence, belief and justification?

• Why might experts sometimes be better calibrated than beginners?

Knowledge Questions

• Is certainty a reliable indicator of knowledge?

• What is the difference between confidence, belief and justification?

• Why might experts sometimes be better calibrated than beginners?

Transfer Example

For example, run this activity with ten general knowledge questions with confidence bands from 50% to 100%. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

• One thing I first treated as evidence was...

- My confidence changed because...
- A better knowledge question I can now ask is...